

Math 140 Spring 2023 - Final Prep

Section:

Name:

1. (Spring 2019) $\lim_{x \rightarrow \infty} (\sqrt{x^2 - x - 1} - x)$

2. (Spring 2022) $\lim_{x \rightarrow 0} \frac{\sin(3x^2)}{7x^2}$

3. (Fall 2021) Consider the equation $xy^2 = 18$, with y a differentiable function of x for $x > 0$. Find an equation for the line L tangent to the graph of the equation at the point $(2, -3)$.

4. (Fall 2019) A substance grows exponentially, and its weight doubles every 3 hours. Suppose it weighs 2 pounds at noon. After how many hours will it weigh 9 pounds?

5. (Spring 2019) Find the indefinite integral $\int \frac{1}{\sqrt{x}(1+2\sqrt{x})} dx$.

6. (Exam 4) Use logarithmic differentiation to derive $f(x) = \sqrt[5]{\frac{e^{6x}(x^2+10)^{10}}{x^3(x^4+10)}} \cdot \sqrt[9]{x}$.

7. (Fall 2021) The graph of the equation $9x^2 + 4y^2 - 36x + 8y + 4 = 0$ is a conic section in the xy -plane. Identify the center and the vertices of the conic section, and sketch the conic section with appropriate labels.